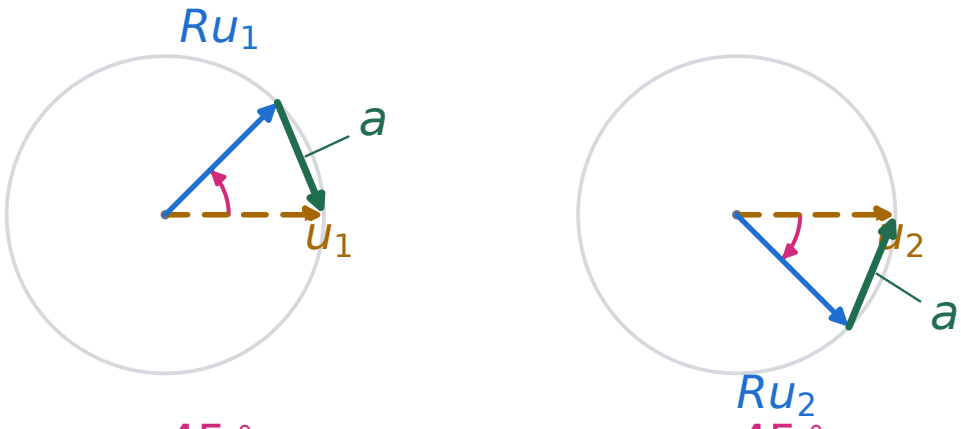
$$U = \mathrm{span}\{e_0, e_1\}, \quad V = R(U) = W(U), \quad \theta_1 = \theta_2 = 45^\circ$$

Direct rotation R

$\Pi_1 \oplus \Pi_2 = \mathbb{R}^4$ (principal planes)

$\Pi_1 = \mathrm{span}\{e_0, e_3\}$ $\quad$ $\Pi_2 = \mathrm{span}\{e_1, e_2\}$

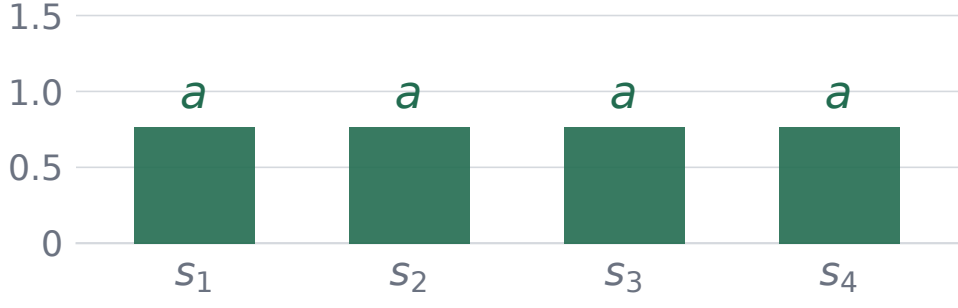


$+45^\circ$ $\quad$ -45°

Unit input q $\quad$ Image Tq

Singular values of $I - R$

$a = \sqrt{2 - \sqrt{2}} \approx 0.765$



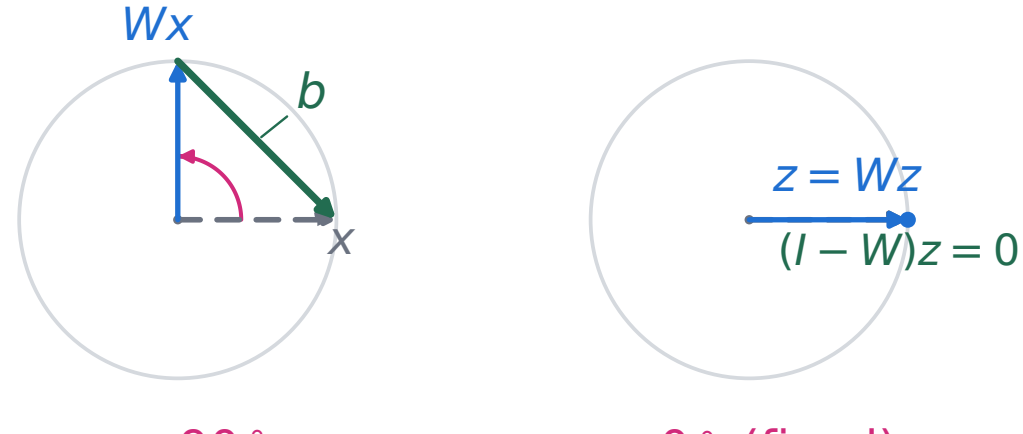
from Π_1 $\quad$ from Π_2

$$\|I - R\|_1 = 4a \approx 3.061$$

Competitor W

$M \oplus Z = \mathbb{R}^4$ (moving and fixed planes)

$M = \mathrm{range}(W - I)$ $\quad$ $Z = \ker(W - I)$

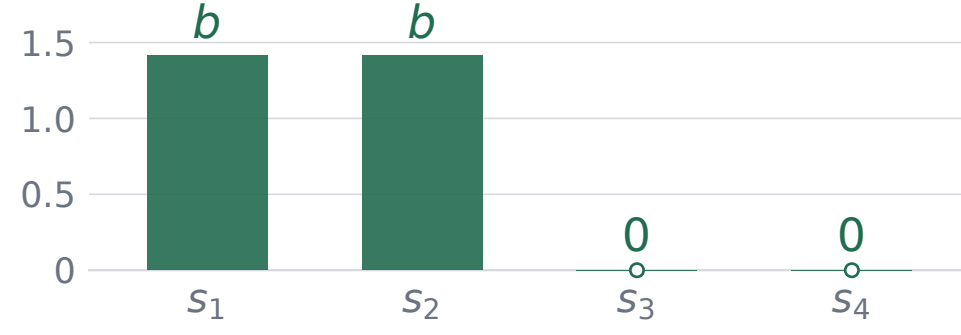


$+90^\circ$ $\quad$ 0° (fixed)

Displacement $(I - T)q$

Singular values of $I - W$

$b = \sqrt{2} \approx 1.414$



from M $\quad$ from Z

$$\|I - W\|_1 = 2b \approx 2.828$$

$$>$$